

CME FEEDER CATTLE FUTURES (GF)

Spread & Curve Structure -- Fundamental Reference

The Stocker-to-Feedlot Pipeline, the Feed Cost Linkage, and the CME Feeder Cattle Index

This document presents a fact-based reference on the fundamental drivers of CME Feeder Cattle futures (ticker: GF), contract size 50,000 lb, the futures market pricing the lighter-weight calves and yearlings that have not yet entered the feedlot finishing stage priced by the Live Cattle contract (covered in a separate reference document in this series). Feeder Cattle occupies a structurally distinct position in the cattle production chain: where Live Cattle prices the finished, feedlot-ready animal destined for slaughter, Feeder Cattle prices the intermediate-stage animal -- typically a weaned calf or yearling weighing roughly 600-900 lb -- that stocker operators and feedlots purchase to place on feed for the 5-7 month finishing period described in the Live Cattle reference document's discussion of the biological production lag. This makes Feeder Cattle the contract most directly exposed to two forces with no equivalent in the Live Cattle market specifically: feed grain costs (since feeder cattle value is mechanically a function of expected finishing costs) and the forage/grazing conditions that determine calf weight gain prior to feedlot placement. This document presents the feeder cattle-specific fundamentals in full, with particular emphasis on the cattle cycle's feeder-specific stage, the corn-feeder cattle price relationship, the CME Feeder Cattle Index settlement mechanism, and the spread relationship to Live Cattle.

Futures First -- Internal Education Material
For Analyst Group Use

1. Contract Specifications and Settlement Mechanics

CME Feeder Cattle is distinguished from Live Cattle by both its underlying animal specification and, critically, its settlement mechanism: unlike Live Cattle's physical delivery system, Feeder Cattle settles in cash against an index, reflecting the practical difficulty of standardising physical delivery for an animal specification (weight, age, breed composition) that varies considerably more than the finished, feedlot-ready animal priced by Live Cattle.

Parameter	Detail
Exchange	CME Group (Chicago Mercantile Exchange)
Ticker	GF (front month)
Underlying	Feeder cattle -- medium and large frame steers, typically 600-900 lb, suitable for feedlot placement
Contract size	50,000 lb (approximately 55-65 head, depending on average individual animal weight)
Quotation	US cents per pound (¢/lb); minimum tick = 0.025¢/lb = \$12.50 per contract
Contract months	January (F), March (H), April (J), May (K), August (Q), September (U), October (V) -- a seven-month annual cycle distinct from Live Cattle's six-month cycle (Feb/Apr/Jun/Aug/Oct/Dec), reflecting the somewhat different seasonal placement and marketing pattern for feeder-weight cattle versus finished cattle.
Settlement type	Cash-settled against the CME Feeder Cattle Index -- a key structural difference from Live Cattle's physical delivery mechanism, addressed in full in Section 1.1.
Last trading day	Last Thursday of the contract month
Contract value at \$2.40/lb	\$120,000 per contract
Relationship to Live Cattle	Feeder Cattle (GF) prices the input to the feedlot finishing process priced by Live Cattle (LE); the two contracts are connected via the feedlot placement and feeding-cost economics described in Section 2.

1.1 Cash Settlement Against the CME Feeder Cattle Index

- **Why cash settlement rather than physical delivery:** feeder cattle vary considerably in weight, frame size, breed composition, and health/management history in a way that makes standardising a single physical delivery specification impractical at the scale and liquidity the futures market requires -- in contrast to Live Cattle, where the finished, feedlot-ready animal is a more standardised commodity closer to its point of slaughter. CME addressed this by settling GF in cash against a published index rather than requiring physical delivery.
- **The CME Feeder Cattle Index construction:** calculated as a 7-day weighted average of feeder cattle prices reported by USDA Agricultural Marketing Service (AMS) across major feeder cattle marketing regions and auction markets, covering a defined weight range and frame/muscle grade specification intended to track the broad feeder cattle cash market rather than any single delivery point.
- **Practical implication for the front contract:** because GF converges to a broad cash-market index rather than to a single delivery location's price, front-month GF basis behaviour and convergence dynamics differ somewhat from Live Cattle's physical-delivery-anchored basis, though the general principle that the front contract increasingly reflects current cash market conditions as expiry approaches still applies.

2. The Feed Cost Linkage -- Why Feeder Cattle Value Is Forward-Looking

The single most important conceptual feature distinguishing Feeder Cattle from Live Cattle is that a feeder animal's value is mechanically determined by the market's expectation of what it will cost to finish that animal to slaughter weight and what the finished animal will be worth -- making Feeder Cattle a forward-looking derivative of both feed grain costs and expected future Live Cattle prices, rather than a market with its own independent value driver.

2.1 The "Cost of Gain" Concept

- **What cost of gain means:** the cost, per pound of weight gained, of finishing a feeder animal to slaughter weight in a feedlot -- determined predominantly by the cost of feed grain (corn, described in the Corn reference document in this series) and protein supplement (soybean meal, described in the Soybean Meal reference document), plus a smaller contribution from yardage (feedlot operating costs) and animal health/management costs.
- **The mechanical price relationship:** a feedlot operator's maximum economically justified bid for a feeder animal is, broadly, the expected finished (slaughter-weight) value of that animal minus the expected cost of gain to get it there minus a required margin -- meaning feeder cattle prices respond inversely to corn and soybean meal prices, all else equal: rising feed costs reduce the price feedlots can profitably pay for feeder cattle, and vice versa.
- **The "feeding margin" or "cattle feeding margin":** the calculated profitability of placing a feeder animal on feed at current feeder cattle, corn, and soybean meal prices, given an expected finished cattle (Live Cattle) sale price at the anticipated marketing date -- the cattle-sector equivalent of the soybean crush margin (described in the CBOT Soybeans reference document) or the ethanol crush margin (described in the Corn reference document), and similarly both a real-world feedlot hedging reference and, at times, a directly monitored cross-market relationship for speculative participants.

2.2 The Corn-Feeder Cattle Relationship

- Because corn is the dominant feed grain cost component described in Section 2.1, the corn-feeder cattle price relationship is one of the most consistently observed cross-market relationships in the agricultural futures complex: sustained corn price increases have historically been associated with feeder cattle price weakness (as feedlot finishing economics deteriorate, reducing feedlot demand for feeder cattle and pressuring feeder cattle prices lower), while sustained corn price declines have historically supported feeder cattle prices via the opposite mechanism.
- **The 2012 drought as a documented illustration:** the severe 2012 US Corn Belt drought described in the Corn reference document, which drove corn prices to record nominal highs, coincided with significant pressure on feeder cattle prices and feedlot profitability, as the sharply higher cost of gain reduced what feedlots could profitably bid for feeder cattle even as drought-driven herd liquidation (Section 4) was separately affecting the supply side of the feeder market.

3. Production Stages -- Where Feeder Cattle Sits in the Pipeline

Understanding Feeder Cattle requires placing it precisely within the multi-stage US cattle production pipeline described at a high level in the Live Cattle reference document.

Stage	Animal Description	Typical Duration	Relevant Futures Contract	Key Activity
Cow-calf	Breeding cow herd produces calves; calves nurse and graze with the cow-calf herd	Calf born to weaning, ~6-8 months	No direct futures contract -- this stage is upstream of both GF and LE	Calving, branding, weaning; the foundational stage determining national calf crop size
Stocker / backgrounding	Weaned calves grazing on pasture or fed a forage-based ration to add frame and weight before feedlot entry	Weaning to feedlot placement, ~3-9 months depending on forage availability and target placement weight	GF prices animals at and around this stage and the point of feedlot placement	Weight gain via grazing/forage; this stage is where pasture and forage conditions (Section 5) most directly affect cattle economics
Feedlot finishing	Feeder-weight animal placed on a grain-based finishing ration in a commercial feedlot	Feedlot placement to slaughter weight, ~5-7 months (the lag described in the Live Cattle reference document)	LE prices the finished animal at the end of this stage	Grain-based weight gain to slaughter weight; this stage is where the feed cost linkage described in Section 2 operates directly

3.1 Why GF Prices the Stocker/Placement Stage Specifically

- GF's underlying specification (medium and large frame steers, typically 600-900 lb) corresponds to animals at or near the point of feedlot placement -- meaning GF prices reflect the market's assessment of feeder cattle value at exactly the transition point between the stocker/backgrounding stage and the feedlot finishing stage, making it the natural hedging instrument for both stocker operators selling animals into feedlots and feedlots purchasing animals for placement.

4. The Cattle Cycle From the Feeder Side

The multi-year cattle cycle described in the Live Cattle reference document -- the biological lag between breeding decisions and slaughter-ready supply -- has a feeder-specific manifestation that is, in some respects, an earlier and more direct signal of the cycle's position than Live Cattle futures themselves.

4.1 The National Calf Crop as the Feeder Supply Foundation

- **USDA Cattle Inventory report:** published semi-annually (January and July) by USDA NASS, providing the national beef cow herd size and the calf crop estimate -- the foundational data determining the total feeder cattle supply available over the following 1-2 years, since the calf crop born in a given year becomes the feeder cattle supply weaned and placed on feed over the subsequent 6-18 months.
- **Herd expansion vs liquidation phases:** as described in the Live Cattle reference document's discussion of the multi-year cattle cycle, when cow herd numbers are being rebuilt (following a period of low cattle prices or recovery from drought-driven liquidation), ranchers retain a larger share of heifers for breeding rather than selling them as feeder cattle -- temporarily reducing feeder cattle supply available to the market even as the longer-run herd rebuild eventually produces a larger calf crop and more feeder cattle supply in subsequent years.
- **Drought-driven liquidation as a feeder-supply-increasing, then feeder-supply-decreasing event:** a severe drought reducing pasture and forage availability (Section 5) can force ranchers to sell breeding stock and calves earlier and in greater volume than they otherwise would -- a short-term increase in feeder cattle supply reaching the market (bearish for GF at the time) that is followed, in subsequent years, by a smaller calf crop and reduced feeder cattle supply as the breeding herd has been reduced (bullish for GF in the following 1-3 years).

4.2 Heifer Retention as a Leading Indicator

- The share of heifers retained for breeding versus sold into the feeder/slaughter channel, tracked via USDA Cattle Inventory report sub-components and industry analysis, is a closely watched leading indicator of whether the cattle cycle is in an expansion phase (rising heifer retention, signalling herd rebuilding and a future increase in calf crop and feeder supply) or a liquidation/maintenance phase (falling or stable heifer retention) -- providing an earlier read on the multi-year supply trajectory than waiting for the calf crop itself to materialise.

5. Pasture and Forage Conditions -- A Feeder-Specific Supply Variable

Because the stocker/backgrounding stage (Section 3) relies substantially on grazing and forage rather than purchased grain, pasture and forage conditions are a feeder-cattle-specific fundamental variable with a more direct and immediate effect on GF than on LE.

5.1 Drought and Pasture Conditions

- **USDA/NASS pasture and range condition ratings:** published weekly during the growing season alongside the broader Crop Progress report (described in the Corn reference document), rating pasture and rangeland condition on the same Excellent/Good/Fair/Poor/Very Poor scale used for row crops -- a direct read on the forage availability determining stocker cattle weight gain economics and grazing capacity in major cattle-producing regions (the Southern Plains, the Flint Hills of Kansas, and other significant grazing regions).
- **The drought-to-feeder-supply mechanism:** as described in Section 4.1, poor pasture conditions force earlier-than-planned movement of cattle off pasture and into either the feeder market (for sale to feedlots, often at lighter-than-ideal weights given reduced grazing-stage weight gain) or directly into feedlots at younger ages -- a documented short-term supply increase that has, at various points, pressured GF prices even while signalling a longer-run supply reduction via the herd-liquidation channel described in Section 4.1.

5.2 Wheat Pasture -- A Region-Specific Forage Source

- In parts of the Southern Plains (particularly Oklahoma, Kansas, and Texas), winter wheat fields are, in some years and operations, grazed by stocker cattle during the autumn-to-spring period before the wheat's own grain-harvest cycle resumes in earnest (the "graze-out" or dual-purpose wheat system) -- connecting regional winter wheat planting and early-season growing conditions (described in the Wheat reference document) to regional stocker cattle grazing capacity and, by extension, to feeder cattle supply and weight-gain economics in this specific region.

6. The Forward Curve Structure

GF's forward curve reflects the combined influence of the cattle cycle's feeder-specific supply dynamics (Section 4), seasonal placement and marketing patterns, and the feed cost linkage (Section 2), which connects GF curve pricing to the corn forward curve.

6.1 Seasonal Placement Patterns

- **Spring and autumn as the two major placement seasons:** feeder cattle placement into feedlots shows a documented seasonal pattern with two broad peaks -- a spring placement season (as winter-grazed cattle, including wheat-pasture cattle described in Section 5.2, come off pasture) and an autumn placement season (as spring/summer-grazed cattle reach placement weight before winter) -- creating a recurring seasonal pattern in the supply of feeder cattle reaching the market.
- **Interaction with the GF contract month calendar:** the seven-month GF contract cycle (Section 1) is structured to provide hedging coverage around these placement-season supply patterns, with contract month selection by hedgers (stocker operators selling into the placement season, feedlots buying for placement) reflecting the specific timing of their own physical activity.

6.2 The Feed Cost Curve Linkage

- Because feeder cattle value is mechanically tied to expected cost of gain (Section 2.1), the GF forward curve is influenced by the CBOT Corn forward curve (described in the Corn reference document) at the corresponding future date -- a feeder contract maturing into a period when the corn curve implies higher expected feed costs will, all else equal, be pressured relative to a feeder contract maturing into a period of expected lower feed costs.

6.3 The Live Cattle-Feeder Cattle Spread

- The spread between Live Cattle and Feeder Cattle prices (adjusted for the different contract units and weight bases) is a direct read on feedlot finishing margin expectations -- a widening LE-GF spread (Live Cattle expensive relative to Feeder Cattle) signals improving expected feedlot profitability and, mechanically, supports feedlot demand for feeder cattle at the margin, while a narrowing spread signals the opposite. This relationship is closely related to, but distinct from, the full cattle feeding margin calculation described in Section 2.1, which also incorporates the corn and soybean meal cost inputs explicitly.

7. Macro and Cross-Market Drivers

7.1 Corn and Soybean Meal -- The Primary Cost Inputs

- As described in Section 2, corn (the energy/caloric feed component, described in the Corn reference document) and soybean meal (the protein component, described in the Soybean Meal reference document) are the dominant feed cost inputs determining cost of gain, making both the CBOT corn and soybean meal markets direct, continuously relevant cross-market inputs for GF -- more directly and mechanically relevant to GF than to LE, since LE prices an animal that has already been fed to completion rather than one whose remaining feeding cost is still being priced in.

7.2 Live Cattle as a Forward Demand Signal

- Because feeder cattle value depends on the expected finished-animal sale price (Section 2.1), Live Cattle futures prices for the contract month corresponding to a feeder animal's expected marketing date are a direct input into feeder cattle valuation -- meaning GF curve pricing should be assessed jointly with the corresponding LE contract rather than purely on feeder-specific supply data, particularly for the back end of the GF curve where the relevant LE marketing date is further in the future.

7.3 Speculative Positioning

- **CFTC COT for CME Feeder Cattle:** weekly managed money net positioning is tracked, consistent with the positioning framework described throughout this document series, and is frequently monitored alongside the related Live Cattle and corn positioning data given the cost-of-gain and finishing-margin relationships connecting all three markets.

8. Documented Seasonal Patterns and Historical Events

8.1 Seasonal Patterns

- **Spring and autumn placement seasons:** as described in Section 6.1, feeder cattle supply reaching the market shows two broad seasonal peaks tied to the spring and autumn placement windows, creating a recurring seasonal pattern in GF basis and curve shape.
- **Wheat pasture graze-out timing:** in the Southern Plains specifically (Section 5.2), the timing of wheat pasture graze-out (cattle coming off wheat fields as the wheat's own growing season resumes in spring) creates a regionally specific seasonal supply pattern layered on top of the broader national placement seasonality.

8.2 Documented Historical Events

- **2012 US drought:** as described in Section 2.2, the severe 2012 Corn Belt drought simultaneously drove corn prices to record highs (pressuring feeding margins and feeder cattle demand) and degraded pasture conditions across cattle-producing regions (forcing earlier-than-planned cattle movement off pasture, per Section 5.1) -- a compounding effect illustrating how a single weather event can affect feeder cattle fundamentals through multiple distinct channels simultaneously.
- **Multi-year drought-driven herd liquidation (various episodes):** sustained drought across major cattle-producing regions, including notable episodes affecting Texas and the broader Southern Plains, has at various points driven the US beef cow herd to multi-decade lows, with the herd-liquidation-then-rebuild cycle described in Section 4.1 working through feeder cattle supply over the following several years.

9. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Typical Timing	Primary GF Curve Impact
USDA Cattle Inventory Report	USDA NASS	Semi-annual	Late January and late July	All contracts -- national beef cow herd and calf crop, the foundational feeder supply data
USDA Cattle on Feed Report	USDA NASS	Monthly	3rd or 4th Friday, ~3:00 PM ET	Front-to-mid contracts -- placements data directly relevant to near-term feeder cattle demand from feedlots
CME Feeder Cattle Index	CME Group, sourced from USDA AMS data	Daily	Published after USDA regional reporting	Front contract -- the direct cash-settlement reference
USDA AMS regional feeder cattle auction reports	USDA AMS	Daily/weekly by region	Variable by market	Front contract -- underlying data feeding the CME Feeder Cattle Index
USDA/NASS Crop Progress (pasture and range conditions)	USDA NASS	Weekly (Apr-Nov)	Monday ~16:00 ET	Front-to-mid contracts -- forage availability for stocker/backgrounding weight gain
NOAA/NDMC US Drought Monitor	National Drought Mitigation Center / USDA	Weekly (Thursday)	Continuous	Mid-to-back contracts -- drought-driven herd liquidation risk, the multi-year supply signal
WASDE (Corn component)	USDA WAOB	Monthly	~11th-12th of month, 12:00 ET	All contracts -- feed cost input via the corn balance sheet, see the Corn reference document
CFTC Commitments of Traders (Feeder Cattle)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding -- tracked alongside Live Cattle and corn given the cost-of-gain relationship

CME Feeder Cattle prices the intermediate stage of the US cattle production pipeline -- the stocker/backgrounding-to-feedlot-placement transition -- making it structurally distinct from Live Cattle in two specific respects: its value is mechanically forward-looking, derived from expected feed costs and expected future finished-cattle prices rather than from its own independent value driver, and its supply side is directly exposed to pasture and forage conditions in a way Live Cattle, pricing an animal already committed to grain-based feedlot finishing, is not. The cash-settlement-against-index mechanism, rather than physical delivery, is itself a direct consequence of the greater heterogeneity of the underlying animal specification at this production stage.

For spread traders, the most reliable analytical framework treats GF as sitting at the intersection of three distinct fundamental inputs: the feeder-specific supply side (USDA Cattle Inventory, heifer retention trends, and pasture/drought conditions), the feed cost side (the CBOT corn and soybean meal curves), and the forward demand side (the corresponding-maturity Live Cattle contract, via the cattle feeding margin). The LE-GF spread and the broader cattle feeding margin calculation are the primary cross-market relationships connecting GF to the rest of the cattle and feed-grain complex described elsewhere in this document series.